Create a database named library and following TABLES in the database:

1. Branch
2. Employee
3. Books
4. Customer
5. IssueStatus
6. ReturnStatus

Attributes for the tables:

1. Branch

- Branch_no - Set as PRIMARY KEY
- Manager_Id
- Branch_address
- Contact_no

```sql
CREATE TABLE Branch (
    Branch_no INT PRIMARY KEY,
    Manager_Id INT,
    Branch_address VARCHAR(100),
    Contact_no VARCHAR(15)
);
```

Output

Branch_no	Manager_Id	Branch_address	Contact_no
1	101	MG Road, Kochi	9876543210
2	102	Trivandrum Central	9876543211
3	103	Calicut Main Branch	9876543212
4	104	Thrissur Town	9876543213
NULL	NULL	NULL	NULL

2. Employee

- Emp_Id – Set as PRIMARY KEY
- Emp_name
- Position
- Salary
- Branch_no - Set as FOREIGN KEY and it refer Branch_no in Branch table

```sql
CREATE TABLE Employee (
    Emp_Id INT PRIMARY KEY,
    Emp_name VARCHAR(50),
    Position VARCHAR(50),
    Salary DECIMAL(10,2),
    Branch_no INT,
    FOREIGN KEY (Branch_no) REFERENCES Branch(Branch_no)
);
```

3. Books

- ISBN - Set as PRIMARY KEY
- Book_title
- Category
- Rental_Price
- Status [Give yes if book available and no if book not available]
- Author
- Publisher

```sql
CREATE TABLE Books (
    ISBN VARCHAR(20) PRIMARY KEY,
    Book_title VARCHAR(100),
    Category VARCHAR(50),
    Rental_Price DECIMAL(10,2),
    Status VARCHAR(5),
    Author VARCHAR(50),
    Publisher VARCHAR(50)
);
```

4. Customer

- Customer_Id - Set as PRIMARY KEY

- Customer_name
- Customer_address
- Reg_date

```sql
CREATE TABLE Customer (
    Customer_Id INT PRIMARY KEY,
    Customer_name VARCHAR(50),
    Customer_address VARCHAR(100),
    Reg_date DATE
);
```

5. IssueStatus

- Issue_Id - Set as PRIMARY KEY
- Issued_cust – Set as FOREIGN KEY and it refer customer_id in CUSTOMER table
  Issued_book_name
- Issue_date
- Isbn_book – Set as FOREIGN KEY and it should refer isbn in BOOKS table

```sql
CREATE TABLE IssueStatus (
    Issue_Id INT PRIMARY KEY,
    Issued_cust INT,
    Issued_book_name VARCHAR(100),
    Issue_date DATE,
    Isbn_book VARCHAR(20),
    FOREIGN KEY (Issued_cust) REFERENCES Customer(Customer_Id)
    FOREIGN KEY (Isbn_book) REFERENCES Books(ISBN)
);
```

6. ReturnStatus

- Return_Id - Set as PRIMARY KEY
- Return_cust
- Return_book_name
- Return_date

- Isbn_book2 - Set as FOREIGN KEY and it should refer isbn in BOOKS table

```sql
CREATE TABLE ReturnStatus (
    Return_Id INT PRIMARY KEY,
    Return_cust INT,
    Return_book_name VARCHAR(100),
    Return_date DATE,
    Isbn_book2 VARCHAR(20),
    FOREIGN KEY (Isbn_book2) REFERENCES Books(ISBN)
);
```

1. Retrieve the book title, category, and rental price of all available books.

```sql
SELECT Book_title, Category, Rental_Price
FROM Books
WHERE Status = 'yes';
```

Output:

Book_title	Category	Rental_Price
World History	History	30.00
Modern Physics	Science	25.00
Database Systems	Technology	40.00
English Literature	Literature	20.00
Ancient History	History	32.00

2. List the employee names and their respective salaries in descending order of salary.

- ```sql
  SELECT Emp_name, Salary
  FROM Employee
  ORDER BY Salary DESC;
  ```
  ```

Output:

Emp_name	Salary
Arun Kumar	75000.00
Meera Nair	72000.00
Rahul Das	70000.00
Anita Joseph	68000.00
Divya K	42000.00
Neha Paul	40000.00
Nithin B	39000.00
Sneha M	38000.00
Vishnu Raj	35000.00
Kiran V	33000.00
Manu P	32000.00
Ravi T	31000.00
Akhil S	30000.00

3. Retrieve the book titles and the corresponding customers who have issued those books.

```
SELECT IssueStatus.Issued_book_name, Customer.Customer_name
FROM IssueStatus
JOIN Customer
ON IssueStatus.Issued_cust = Customer.Customer_Id;
```

Output:

Issued_book_name	Customer_name
World History	Ajay Menon
Database Systems	Priya Nair
English Literature	Anjali Das
Ancient History	Rahul Pillai

4. Display the total count of books in each category.

```
SELECT Category, COUNT(*) AS Total_Books
FROM Books
GROUP BY Category;
```

Output:

Category	Total_Books
History	3
Science	1
Technology	2
Literature	1

5. Retrieve the employee names and their positions for the employees whose salaries are above Rs.50,000.

```
SELECT Emp_name, Position
FROM Employee
WHERE Salary > 50000;
```

Output:

Emp_name	Position
Arun Kumar	Manager
Meera Nair	Manager
Rahul Das	Manager
Anita Joseph	Manager

6. List the customer names who registered before 2022-01-01 and have not issued any books yet.

```
SELECT Customer_name
FROM Customer
WHERE Reg_date < '2022-01-01'
AND Customer_Id NOT IN
(SELECT Issued_cust FROM IssueStatus);
```

Output:

Customer_name
Suresh Babu

7. Display the branch numbers and the total count of employees in each branch.

```
SELECT Branch_no, COUNT(*) AS Total_Employees
FROM Employee
GROUP BY Branch_no;
```

Output:

	Branch_no	Total_Employees
▶	1	3
	2	3
	3	6
	4	1

8. Display the names of customers who have issued books in the month of June 2023.

```
• SELECT DISTINCT Customer.Customer_name
  FROM IssueStatus
  JOIN Customer
  ON IssueStatus.Issued_cust = Customer.Customer_Id
  WHERE MONTH(Issue_date) = 6
  AND YEAR(Issue_date) = 2023;
```

Output:

	Customer_name
▶	Ajay Menon
	Priya Nair
	Rahul Pillai

9. Retrieve book_title from book table containing history.

```
SELECT Book_title
FROM Books
WHERE Book_title LIKE '%history%';
```

Output:

	Book_title
▶	World History
	Indian History
	Ancient History

10.Retrieve the branch numbers along with the count of employees for branches having more than 5 employees

```
SELECT Branch_no, COUNT(*) AS Employee_Count
FROM Employee
GROUP BY Branch_no
HAVING COUNT(*) > 5;
```

Output:

	Branch_no	Employee_Count
▶	3	6

11. Retrieve the names of employees who manage branches and their respective branch addresses.

```
SELECT Employee.Emp_name, Branch.Branch_address
FROM Employee
JOIN Branch
ON Employee.Emp_Id = Branch.Manager_Id;
```

Output:

	Emp_name	Branch_address
▶	Arun Kumar	MG Road, Kochi
	Meera Nair	Trivandrum Central
	Rahul Das	Calicut Main Branch
	Anita Joseph	Thrissur Town

12. Display the names of customers who have issued books with a rental price higher than Rs. 25.

```
SELECT DISTINCT Customer.Customer_name
FROM Customer
JOIN IssueStatus
ON Customer.Customer_Id = IssueStatus.Issued_cust
JOIN Books
ON IssueStatus.Isbn_book = Books.ISBN
WHERE Books.Rental_Price > 25;
```

Output:

	Customer_name
▶	Ajay Menon
	Priya Nair
	Rahul Pillai